0.5cm 0cm
@,@*)

JavaScript Course

Source Code Collection

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Code Academy | SEA Technologies Media

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Code Error Legend

Errors Found in Source Code

The following errors were detected in the original code and have been flagged in-place with corrections alongside. Both the **incorrect** (red) and **correct** (green) versions appear so learners can compare.

Error 1 — Operators: `console.log|(a === b);` contains a stray pipe character `|` that makes it invalid syntax.
Correct: `console.log(a === b);`

Error 2 — Loops (forEach): `console.log('${index + 1}. ${course}');` uses single quotes instead of backticks, so template literals will not interpolate.
Correct: use backticks `` `${index + 1}. ${course}` ``

Error 3 — Loops (while): `console.log("Count;" + count);` has a semicolon inside the string instead of a colon.
Correct: `console.log("Count:" + count);`

Error 4 — Loops (for): `console.log("Lesson" + i)` missing a space before the number, output will be `Lesson1` instead of `Lesson 1`.
Correct: `console.log("Lesson " + i)`

Error 5 — Arrays: `coursesInCodeAacdemy` typo: `Aacdemy` should be `Academy`.
Correct: `coursesInCodeAcademy`

Error 6 — Objects: `students.phone = "09055374715",` ends with a comma but should end with a semicolon.
Correct: `students.phone = "09055374715";`

Error 7 — Comments: `//LOgical Operators` typo: `LOgical` should be `Logical`, and missing space after `//`.
Correct: `// Logical Operators`

Error 8 — Comments: `//Condtions with logical operators` typo: `Condtions` should be `Conditions`, and missing space after `//`.
Correct: `// Conditions with logical operators`

Error 9 — Comments: `//While loop __ runs while conditionis true` typo: `conditionis` should be `condition is`.
Correct: `// While loop --- runs while condition is true`

Error 10 — Style: `console.log ("name" in students);` space before `(` is inconsistent with JavaScript style conventions.
Preferred: `console.log("name" in students);`

SEA TE

Topic 1: What is JavaScript and How It Works — index_html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>JavaScript Basics</title>
</head>
<body>
    <h1>Welcome to JavaScript Basics</h1>
    <p>This page demonstrates basic JavaScript functionality.</p>
    <button onclick="showAlert)">Click Me!</button><!-- Internal JavaScript --><!-- <script>function showAlert() alert("Hello,
     World!");console.log("Internal JavaScript is working!");</script> --><script src="script.js"></script></body></html>
```

Topic 1: What is JavaScript and How It Works — script_js

```
// console.log print to the browser console


console.log"External JS is working");console.log("Welcome to Code Academy!");//alert displays a pop-up message in the browser//alert shows a
     pop-up to the useralert("Welcome to Code Academy!");// A simple functionfunction sayHello() alert("Hello! Javascript is working!");
```

SEA TECHNOLOGIES

Topic 2: Variables and Data Types

```
//------ VARIABLES ------

// const - value cannnot be changed


const name = "Emmanuel Adedayo";
const platformName = "Code Academy";
const year = 2026;

// let - value can be changed


let score = 0;
let isLoggedIn = false;
let currentLesson = "JavaScript Basics";

//Changing a Let variable


score = 100;
currentLesson = "JavaScript Functions";
console.logscore); //100console.log(currentLesson); //JavaScript Functions// name = "Emmanuel"; // This will throw an error because name is
      a const variable//-------DATA TYPES ------// Stringconst firstName = "Emmanuel";const lastName = "Adedayo";const course =
      'JavaScript';const greeting = `Hello, $firstName $lastName! Welcome to $course`; // Template literal// Numberconst age = 25;const
      price = 19.99;const percentage = 0.75;// Booleanconst isFree = true;const isPaid = false;const isActive = true;//Undefined -- variable
      declared but not assigned a valuelet myVariable;console.log(myVariable); // undefined//Null -- variable explicitly set to have no
      valueconst emptyValue = null;//Array -- List of mutiple itemsconst courses = ["HTML", "CSS", "JavaScript", "Python", "Git",
      "Django"];//Object -- Collection of key-value pairsconst student = name: "Emmanuel Adedayo",age: 25,course: "JavaScript",isActive:
      true;//-----CHECKING DATA TYPES-----console.log(typeof firstName); // stringconsole.log(typeof age); // numberconsole.log(typeof
      isFree); // booleanconsole.log(typeof myVariable); // undefinedconsole.log(typeof courses); // objectconsole.log(typeof student); //
      object// ---TEMPLATE LITERALS---//Old way of concatenating strings +const oldWay = "Hello, " + firstName + " " + lastName + "! Welcome
      to " + course + ".";//---New way using template literals with backticks and $ for variablesconst newWay = `Hello, $firstName
      $lastName! Welcome to $course.`;console.log(oldWay);console.log(newWay);
```

SEA TECHNOLOGIES

Topic 3: Operators

```
//---OPERATORS---

// Arithmetic Operators


const a = 20;
const b = 10;

console.loga + b); // 30 - Additionconsole.log(a - b); // 10 - Subtractionconsole.log(a  b);  // 200 - Multiplication
console.loga / b); // 2 - Divisionconsole.log(a console.log(a * b); // 10240000000000 - Exponentiation

//Comparison Operators


console.loga > b); // true - Greater thanconsole.log(a < b); // false - Less thanconsole.log|(a === b);
      (@\textcolor{errorflag}{\bfseries ERROR: remove |} @*)
console.loga === b); (@\textcolor{correctclr}{\bfseries CORRECT} @*) // false - Strict equality
console.loga !== b); // true - Strict inequalityconsole.log(a >= b); // true - Greater than or equal toconsole.log(a <= b); // false - Less
      than or equal to//LOgical Operators (@\textcolor{errorflag}{\bfseries TYPO + space} @*)
// Logical Operators  @\textcolor{correctclr}{\bfseries CORRECT} @*)


console.logtrue && true); // true - ANDconsole.log(true && false); // false - ANDconsole.log(false && true); // false - ANDconsole.log(false
      && false); // false - ANDconsole.log(true || true); // true - ORconsole.log(true || false); // true - ORconsole.log(false || true); //
      true - ORconsole.log(false || false); // false - ORconsole.log(!true); // false - NOTconsole.log(!false); // true - NOT
```

SEA TECHNOLOGIES

Topic 4: Conditions and Loops

```
//---CONDITIONAL STATEMENTS---

//Basic if else statement


const userAge = 18;

if userAge >=18) console.log("You are an Adult and you are eligible to vote."); else console.log("You are not an Adult and you are not
      eligible to vote.");//if else if else statementconst examScore = 85;if (examScore >= 85) console.log("Distinction -- Excellent
      performance!"); else if (examScore >= 70) console.log("Credit -- Good performance!"); else if (examScore >= 50) console.log("Pass --
      Satisfactory performance!"); else if (examScore >= 35) console.log("Fail -- Unsatisfactory performance!"); else console.log("Fail --
      Poor performance!");//Condtions with logical operators (@\textcolor{errorflag}{\bfseries TYPO + space} @*)
// Conditions with logical operators  @\textcolor{correctclr}{\bfseries CORRECT} @*)


const  hasCode = true;
const isRegistered = true;

if hasCode && isRegistered) console.log("Access granted! You can enter the course."); else console.log("Access denied! Please register and
      get your code");//Ternary operator -- short form of if else statementconst status = userAge >= 18 ? "Adult" :
      "Minor";console.log(status); // Adult//Switch statementconst day = "Monday";switch (day) case "Monday":console.log("Today is Monday -
      Start of the week!");break;case "Tuesday":console.log("Today is Tuesday");break;case "Wednesday":console.log("Today is
      Wednesday");break;case "Thursday":console.log("Today is Thursday");break;case "Friday":console.log("Today is Friday - End of the
      week!");break;case "Saturday":case "Sunday":console.log("Weekend");break;default:console.log("Invalid day!");//-----LOOPS------// For
      loops -- runs a specific number of timesfor (let i = 1; i <= 10; i++) console.log("Lesson" + i)
      (@\textcolor{errorflag}{\bfseries ERROR: missing space} @*)
    console.log"Lesson " + i) (@\textcolor{correctclr}{\bfseries CORRECT} @*)
}

// For loop through an array


const allCourses = ["HTML", "CSS", "JavaScript", "Python", "Git", "Django"];

forlet i = 0; i < allCourses.length; i++)console.log(i + "," + allCourses[i]);//forEach -- cleaner way to loop through an
      arrayallCourses.forEach(function(course, index) console.log('$index + 1. $course');
      (@\textcolor{errorflag}{\bfseries ERROR: use backticks} @*)
    console.log`$index + 1. $course`); (@\textcolor{correctclr}{\bfseries CORRECT} @*)
});

//While loop -- runs while conditionis true  @\textcolor{errorflag}{\bfseries TYPO: conditionis} @*)
// While loop --- runs while condition is true  @\textcolor{correctclr}{\bfseries CORRECT} @*)


let count = 1;
while count <= 5) console.log("Count;" + count); (@\textcolor{errorflag}{\bfseries ERROR: use colon} @*)
console.log"Count:" + count); (@\textcolor{correctclr}{\bfseries CORRECT} @*)
    count++
}

//break -- stop loop early


for let i = 0; i < 10; i++) if (i === 5)break;console.log(i)//print 0 1 2 3 4//continue -- skip one iterationfor (let i = 0; i <10; i++) if
      (i === 5) continue;console.log(i)//skip 5, print everything else
```

SEA TECH

Topic 5: Functions

```
// ── REGULAR FUNCTION ──────────────────────

function greet) console.log("Hello and welcome to Code Academy!");// Call the function — nothing happens until you call
      itgreet();greet();greet();// ── FUNCTION WITH PARAMETERS ──────────function greetStudent(name, course) console.log(`Hello $name!
      Welcome to $course.`);greetStudent("Emmanuel", "JavaScript");greetStudent("Dave", "Python");greetStudent("John", "HTML");// ──
      FUNCTION WITH RETURN VALUE ──────────function addNumbers(a, b) return a + b;const result = addNumbers(10, 5);console.log("Result: " +
      result); // 15// Using return value directlyconsole.log(addNumbers(100, 200)); // 300// ── FUNCTION WITH DEFAULT PARAMETERS
      ────function createGreeting(name, platform = "Code Academy") return `Welcome $name to
      $platform!`;console.log(createGreeting("Emmanuel"));// Welcome Emmanuel to Code Academy!console.log(createGreeting("Dave", "SEA
      TECH"));// Welcome Dave to SEA TECH!// ── ARROW FUNCTIONS ──────────────────────// Regular functionfunction multiply(a, b) return a  b;
}

// Same as arrow function


const multiplyArrow = a, b) => return a  b;
};

// Short arrow function — one line return


const square = n) => n  n;
const double = n => n * 2; // brackets optional for one param

console.logmultiply(4, 5)); // 20console.log(multiplyArrow(4, 5)); // 20console.log(square(6)); // 36console.log(double(7)); // 14// ── REAL
      WORLD FUNCTION EXAMPLES ─────────// Calculate gradefunction getGrade(score) if (score >= 70) return "Distinction";if (score >= 50)
      return "Pass";if (score >= 40) return "Nearly Pass";return "Fail";console.log(getGrade(85)); // Distinctionconsole.log(getGrade(55));
      // Passconsole.log(getGrade(30)); // Fail// Format currencyfunction formatCurrency(amount, currency = "NGN") return `$currency
      $amount.toLocaleString()`;console.log(formatCurrency(50000)); // NGN 50,000console.log(formatCurrency(9.99, "USD")); // USD 9.99//
      Check if access code is valid formatfunction isValidCodeFormat(code) return code.startsWith("SEA-") && code.length ===
      9;console.log(isValidCodeFormat("SEA-USCPK")); // trueconsole.log(isValidCodeFormat("invalid")); // false
```

SEA TECHNOLOGIES

Topic 6: Arrays and Objects

```
//----------ARRAYS---------------------------------------

//Creating arrays


const fruits = ["apple", "banana", "orange"];
const numbers = [1, 2, 3, 4, 5,];
const mixed = ["Emmanuel", 20, true, null];
const coursesInCodeAacdemy  @\textcolor{errorflag}{\bfseries TYPO: Aacdemy} @*) =["HTML", "CSS", "JavaScript", "Python", "Git", "Django"];

console.logcoursesInCodeAacdemy (@\textcolor{errorflag}{\bfseries TYPO: Aacdemy} @*)[0]);
console.logcoursesInCodeAacdemy (@\textcolor{errorflag}{\bfseries TYPO: Aacdemy} @*)[1]);
console.logcoursesInCodeAacdemy (@\textcolor{errorflag}{\bfseries TYPO: Aacdemy} @*)[2]);
console.logcoursesInCodeAacdemy (@\textcolor{errorflag}{\bfseries TYPO: Aacdemy} @*)[3]);
console.logcoursesInCodeAacdemy (@\textcolor{errorflag}{\bfseries TYPO: Aacdemy} @*)[4]);
console.logcoursesInCodeAacdemy (@\textcolor{errorflag}{\bfseries TYPO: Aacdemy} @*)[5]);

//Array length


console.logcoursesInCodeAacdemy (@\textcolor{errorflag}{\bfseries TYPO: Aacdemy} @*).length);

//---------------ARRAY METHODS-------------------------


coursesInCodeAacdemy  @\textcolor{errorflag}{\bfseries TYPO: Aacdemy} @*).push"React")console.log(coursesInCodeAacdemy
      (@\textcolor{errorflag}{\bfseries TYPO: Aacdemy} @*))

coursesInCodeAacdemy  @\textcolor{errorflag}{\bfseries TYPO: Aacdemy} @*).pop)console.log(coursesInCodeAacdemy
      (@\textcolor{errorflag}{\bfseries TYPO: Aacdemy} @*))

coursesInCodeAacdemy  @\textcolor{errorflag}{\bfseries TYPO: Aacdemy} @*).unshift"Fundamental")console.log(coursesInCodeAacdemy
      (@\textcolor{errorflag}{\bfseries TYPO: Aacdemy} @*))

coursesInCodeAacdemy  @\textcolor{errorflag}{\bfseries TYPO: Aacdemy} @*).shift)console.log(coursesInCodeAacdemy
      (@\textcolor{errorflag}{\bfseries TYPO: Aacdemy} @*))

console.logcoursesInCodeAacdemy (@\textcolor{errorflag}{\bfseries TYPO: Aacdemy} @*).indexOf"Python"));console.log(coursesInCodeAacdemy
      (@\textcolor{errorflag}{\bfseries TYPO: Aacdemy} @*).includes"Python"));console.log(coursesInCodeAacdemy
      (@\textcolor{errorflag}{\bfseries TYPO: Aacdemy} @*).includes"React"));// ── ARRAY HIGHER ORDER METHODS ──────────const scores = [85,
      92, 78, 65, 90, 55, 88];// map – transform each itemconst doubled = scores.map(score => score  2);
console.logdoubled);// filter – keep only items that matchconst passing = scores.filter(score => score >= 70);console.log(passing); // [85,
      92, 78, 90, 88]// find – get first item that matchesconst firstFail = scores.find(score => score < 70);console.log(firstFail); //
      65// reduce – calculate single value from arrayconst total = scores.reduce((sum, score) => sum + score, 0);const average = total /
      scores.length;console.log(`Total: $total, Average: $average`);// some – check if at least one matchesconsole.log(scores.some(score =>
      score >= 90)); // true// every – check if all matchconsole.log(scores.every(score => score >= 50)); //
      true//----------------OBJECTS-----------------const students = name: "Emmanuel Adedayo",age: 20,course: "Javascript",email:
      "seatechnologiesmedia@gmail.com",isActive: true,scores: [45, 55, 66, 77],;console.log(students.name);console.log(students
      ["course"]);//adding new itemstudents.phone = "09055374715", (@\textcolor{errorflag}{\bfseries ERROR: use semicolon} @*)
students.phone = "09055374715";  @\textcolor{correctclr}{\bfseries CORRECT} @*)
console.logstudents.phone)//Updating a propertystudents.age = 21;console.log(students.age);delete
      students.isActive;console.log(students)console.log ("name" in students); (@\textcolor{errorflag}{\bfseries STYLE: space before
      @*)console.log("name" in students); (*@PREFERRED @*)console.log ("score" in students);
```

SEA TECH

Topic 7: DOM Manipulation — dom_html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
          initial-scale=1.0">
    <title>JavaScript DOM</title>
    <style>
        body { font-family: Arial, sans-serif;
               max-width: 700px; margin: 40px auto;
               padding: 20px; }
        .box { width: 150px; height: 150px;
               background: \#1E56D6;
               border-radius: 12px;
               margin: 20px 0; }
        button { background: \#1E56D6; color: white;
                 padding: 10px 20px; border: none;
                 border-radius: 8px; cursor: pointer;
                 margin: 5px; font-size: 16px; }
        button:hover { background: \#0A1F44; }
        input { padding: 12px; font-size: 16px;
                border: 1px solid \#ddd;
                border-radius: 8px; width: 100%;
                margin: 5px 0; }
        .output { background: \#f0f4ff; padding: 20px;
                  border-radius: 12px; margin: 15px 0;
                  font-size: 18px; min-height: 50px; }
        .card { background: white; border: 1px solid \#ddd;
                border-radius: 12px; padding: 15px;
                margin: 10px 0; }
    </style>
</head>
<body>

    <h1>DOM Manipulation</h1>

    <!-- Section 1: Finding and changing elements -->
    <h2>1. Finding Elements</h2>
    <p id="myParagraph">This is the original text.</p>
    <p class="highlight">First highlighted paragraph.</p>
    <p class="highlight">Second highlighted paragraph.</p>
    <button onclick="changeText)">Change Text</button><button onclick="changeAll()">Change All</button><!-- Section 2: Changing styles
     --><h2>2. Changing Styles</h2><div class="box" id="colorBox"></div><button onclick="changeColor()">Change Color</button><button
     onclick="hideBox()">Hide</button><button onclick="showBox()">Show</button><button onclick="toggleBox()">Toggle</button><!-- Section 3:
     Creating elements --><h2>3. Creating Elements</h2><div id="courseList"><h3>My Courses:</h3></div><button onclick="addCourse()">Add
     Course</button><button onclick="clearCourses()">Clear All</button><!-- Section 4: Events --><h2>4. Events</h2><input type="text"
     id="nameInput"placeholder="Type your name..."><div class="output" id="nameOutput">Your name appears here...</div><!-- Section 5:
     Counter --><h2>5. Counter</h2><div class="output" id="counter"style="text-align:center; font-size:48px;font-weight:bold;
     color:#1E56D6;">0</div><button id="increaseBtn">+ Increase</button><button id="decreaseBtn">- Decrease</button><button
     id="resetBtn">Reset</button><script src="scripts.js"></script></body></html>
```

Topic 7: DOM Manipulation — scripts_js

```
// ── FINDING ELEMENTS ──────────────────────

// By ID – returns single element


const myParagraph = document.getElementById"myParagraph");// By class – returns all matching elements (HTMLCollection)const highlights =
      document.getElementsByClassName("highlight");// querySelector – returns FIRST matching elementconst firstHighlight =
      document.querySelector(".highlight");// querySelectorAll – returns ALL matching elements (NodeList)const allHighlights =
      document.querySelectorAll(".highlight");console.log(myParagraph);console.log(allHighlights.length); // 2// ── CHANGING CONTENT
      ──────────────────────function changeText() // textContent – changes text onlymyParagraph.textContent = "Text changed by
      JavaScript!";myParagraph.style.color = "#1E56D6";myParagraph.style.fontWeight = "bold";function changeAll() // innerHTML – can include
      HTML tagsallHighlights.forEach((el, i) => el.innerHTML = `Paragraph $i + 1 changed!<strong>Bold text too!</strong>`;el.style.color =
      "#E38A2C";});// ── CHANGING STYLES ──────────────────────const colorBox = document.getElementById("colorBox");function changeColor()
      const colors = ["#1E56D6", "#E38A2C","#1F9D55", "#D64545", "#9B59B6"];const random = colors[Math.floor(Math.random()  colors.length)];
    colorBox.style.backgroundColor = random;
    colorBox.style.borderRadius = "50%"; // make it circle
}

function hideBox) colorBox.style.display = "none";function showBox() colorBox.style.display = "block";colorBox.style.borderRadius =
      "12px";function toggleBox() // classList toggle adds class if missing, removes if presentcolorBox.classList.toggle("hidden");if
      (colorBox.style.display === "none") colorBox.style.display = "block"; else colorBox.style.display = "none";// ── CREATING ELEMENTS
      ──────────────────────const courseNames = ["HTML", "CSS", "JavaScript","Python", "Git & GitHub", "Django"];let courseIndex = 0;function
      addCourse() if (courseIndex >= courseNames.length) alert("All courses added!");return;const courseList =
      document.getElementById("courseList");// Create new elementconst card = document.createElement("div");card.classList.add("card");//
      Set contentcard.innerHTML = `<strong>$courseIndex + 1.$courseNames[courseIndex]</strong><span style="color:green; float:right">[check]
      Added</span>`;// Append to pagecourseList.appendChild(card);courseIndex++;function clearCourses() const courseList =
      document.getElementById("courseList");courseList.innerHTML = "<h3>My Courses:</h3>";courseIndex = 0;// ── EVENTS WITH addEventListener
      ──────────// Input event – fires as user typesconst nameInput = document.getElementById("nameInput");const nameOutput =
      document.getElementById("nameOutput");nameInput.addEventListener("input", function() if (nameInput.value.trim() === "")
      nameOutput.textContent = "Your name appears here...";nameOutput.style.color = "#666"; else nameOutput.textContent =`Hello
      $nameInput.value!Welcome to Code Academy! [party]`;nameOutput.style.color = "#1E56D6";);// Focus and blur
      eventsnameInput.addEventListener("focus", function() nameInput.style.borderColor = "#1E56D6";nameInput.style.outline =
      "none";);nameInput.addEventListener("blur", function() nameInput.style.borderColor = "#ddd";);// ── COUNTER WITH EVENTS
      ──────────────────let count = 0;const counterDisplay = document.getElementById("counter");const increaseBtn =
      document.getElementById("increaseBtn");const decreaseBtn = document.getElementById("decreaseBtn");const resetBtn =
      document.getElementById("resetBtn");function updateCounter() counterDisplay.textContent = count;// Change color based on valueif
      (count > 0) counterDisplay.style.color = "#1F9D55"; else if (count < 0) counterDisplay.style.color = "#D64545"; else
      counterDisplay.style.color = "#1E56D6";increaseBtn.addEventListener("click", function()
```

```
        count++;updateCounter(););decreaseBtn.addEventListener("click", function()
        count--;updateCounter(););resetBtn.addEventListener("click", function() count = 0;updateCounter(););// Keyboard event — press + or -
        keysdocument.addEventListener("keydown", function(event) if (event.key === "+") count++;updateCounter(); else if (event.key === "-")
        count--;updateCounter(););
```

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★ Topic 8: Final Project— Interactive Student Dashboard — final_html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
          initial-scale=1.0">
    <title>Student Dashboard — Code Academy</title>
    <style>
        * { margin: 0; padding: 0;
            box-sizing: border-box; }
        body { font-family: Arial, sans-serif;
               background: \#f0f4ff; min-height: 100vh; }
        header { background: \#0A1F44; color: white;
                 padding: 20px 40px;
                 display: flex;
                 justify-content: space-between;
                 align-items: center; }
        header h1 { font-size: 24px; }
        header p { color: \#4D8CFF; font-size: 14px; }
        .container { max-width: 1000px; margin: 0 auto;
                     padding: 30px 20px; }
        .stats-grid { display: grid;
                      grid-template-columns: repeat4,1fr);gap: 15px; margin-bottom: 25px; .stat-card  background: white; padding:
      20px;border-radius: 12px;border-left: 4px solid #1E56D6;text-align: center; .stat-card h3  font-size: 32px; color: #1E56D6; .stat-card
      p  color: #666; font-size: 14px; .controls  display: flex; gap: 10px;margin-bottom: 20px; flex-wrap: wrap; input, select  padding:
      10px 15px; font-size: 16px;border: 1px solid #ddd;border-radius: 8px; flex: 1; button  background: #1E56D6; color: white;padding: 10px
      20px; border: none;border-radius: 8px; cursor: pointer;font-size: 16px; white-space: nowrap; button:hover  background: #0A1F44;
      .btn-red  background: #D64545; .btn-red:hover  background: #a33; .add-form  background: white; padding: 20px;border-radius:
      12px;margin-bottom: 20px;display: none; .add-form h3  margin-bottom: 15px; .form-row  display: flex; gap: 10px;flex-wrap: wrap;
      .student-card  background: white; padding: 20px;border-radius: 12px;margin-bottom: 10px;display: flex;justify-content:
      space-between;align-items: center;border-left: 4px solid #1E56D6; .student-info h4  color: #0A1F44; .student-info p  color: #666;
      font-size: 14px; .badge  padding: 5px 15px; border-radius: 20px;font-weight: bold; font-size: 14px; .distinction  background: #d4edda;
      color: #155724; .pass  background: #d1ecf1; color: #0c5460; .nearly  background: #fff3cd; color: #856404; .fail  background: #f8d7da;
      color: #721c24; .empty  text-align: center; padding: 60px;color: #666; font-size: 18px; </style></head><body><header><div><h1>Student
      Dashboard</h1><p>Code Academy | SEA TECHNOLOGIES MEDIA</p></div><button onclick="toggleAddForm()">+ Add Student</button></header><div
      class="container"><!-- Stats --><div class="stats-grid"><div class="stat-card"><h3 id="totalStat">0</h3><p>Total
      Students</p></div><div class="stat-card"><h3 id="avgStat">0</h3><p>Average Score</p></div><div class="stat-card"><h3
      id="highStat">0</h3><p>Highest Score</p></div><div class="stat-card"><h3 id="passStat">0</h3><p>Passing Students</p></div></div><!--
      Add Form --><div class="add-form" id="addForm"><h3>Add New Student</h3><div class="form-row"><input type="text"
      id="newName"placeholder="Student name"><input type="number" id="newScore"placeholder="Score (0-100)"min="0" max="100"><input
      type="text" id="newCourse"placeholder="Course"><button onclick="addStudent()">Add</button></div></div><!-- Controls --><div
      class="controls"><input type="text" id="searchInput"placeholder="[search] Search students..."><select id="filterSelect"><option
      value="all">All Grades</option><option value="Distinction">Distinction</option><option value="Pass">Pass</option><option value="Nearly
      Pass">Nearly Pass</option><option value="Fail">Fail</option></select><button onclick="sortByScore()">Sort by Score</button><button
      class="btn-red" onclick="clearAll()">Clear All</button></div><!-- Student List --><div id="studentList"></div></div><script
      src="final.js"></script></body></html>
```

★ Topic 8: Final Project— Interactive Student Dashboard — final_js

```
// ── DATA ──────────────────────────────────

let students = [
    { id: 1, name: "Emmanuel Adedayo",
      score: 85, course: "JavaScript" },
    { id: 2, name: "Dave Thomas",
      score: 92, course: "Python"     },
    { id: 3, name: "John Doe",
      score: 78, course: "HTML"       },
    { id: 4, name: "Jane Smith",
      score: 55, course: "CSS"        },
    { id: 5, name: "Mike Johnson",
      score: 38, course: "Git"        },
    { id: 6, name: "Sarah Williams",
      score: 90, course: "Django"     }
];

let nextId = 7;
let sortAscending = false;

// ── HELPER FUNCTIONS ──────────────────────

function getGradescore) if (score >= 70) return "Distinction";if (score >= 50) return "Pass";if (score >= 40) return "Nearly Pass";return
      "Fail";function getBadgeClass(grade) const classes = "Distinction": "distinction","Pass": "pass","Nearly Pass": "nearly","Fail":
      "fail";return classes[grade] || "pass";// ── STATISTICS ──────────────────────────function updateStats() const total =
      students.length;const scores = students.map(s => s.score);const avg = total > 0? Math.round(scores.reduce((a, b) => a + b, 0) /
      total): 0;const high = total > 0 ? Math.max(...scores) : 0;const passing = students.filter(s => s.score >=
      50).length;document.getElementById("totalStat").textContent = total;document.getElementById("avgStat").textContent =
      avg;document.getElementById("highStat").textContent = high;document.getElementById("passStat").textContent = passing;// ── RENDER
      STUDENTS ──────────────────────function renderStudents(list) const container = document.getElementById("studentList");if (list.length
      === 0) container.innerHTML = `<div class="empty">No students found. Add some students!</div>`;return;container.innerHTML =
      list.map(student => const grade = getGrade(student.score);const badgeClass = getBadgeClass(grade);return `<div
      class="student-card"id="student-$student.id"><div class="student-info"><h4>$student.name</h4><p>Course: $student.course• Score:
      $student.score/100</p></div><div style="display:flex;align-items:center; gap:10px"><span class="badge
      $badgeClass">$grade</span><button class="btn-red"onclick="removeStudent($student.id)"style="padding:6px
      12px;font-size:14px">Remove</button></div></div>`;).join("");// ── ADD STUDENT ──────────────────────────function toggleAddForm()
      const form = document.getElementById("addForm");form.style.display =form.style.display === "none" ||form.style.display === ""? "block"
      : "none";function addStudent() const name = document.getElementById("newName").value.trim();const score =
      parseInt(document.getElementById("newScore").value);const course = document.getElementById("newCourse").value.trim();if (!name ||
      isNaN(score) || !course) alert("Please fill in all fields!");return;if (score < 0 || score > 100) alert("Score must be between 0 and
      100!");return;const newStudent =  id: nextId++, name, score, course ;students.push(newStudent);// Clear
      formdocument.getElementById("newName").value = "";document.getElementById("newScore").value =
```

```
"";document.getElementById("newCourse").value = "";toggleAddForm();updateDisplay();// ── REMOVE STUDENT
──────────────────────function removeStudent(id) if (confirm("Remove this student?")) students = students.filter(s => s.id !==
id);updateDisplay();// ── SEARCH AND FILTER ──────────────────────function getFilteredStudents() const searchTerm =
document.getElementById("searchInput").value.toLowerCase();const filterGrade = document.getElementById("filterSelect").value;return
students.filter(student => const matchesSearch =student.name.toLowerCase().includes(searchTerm)
||student.course.toLowerCase().includes(searchTerm);const matchesGrade =filterGrade === "all" ||getGrade(student.score) ===
filterGrade;return matchesSearch && matchesGrade;);// ── SORT ──────────────────────────────────────function sortByScore() sortAscending =
!sortAscending;students.sort((a, b) =>sortAscending? a.score - b.score: b.score - a.score);updateDisplay();// ── CLEAR ALL
──────────────────────────────function clearAll() if (confirm("Remove all students? This cannot be undone.")) students =
[];updateDisplay();// ── UPDATE DISPLAY ──────────────────────function updateDisplay() const filtered =
getFilteredStudents();renderStudents(filtered);updateStats();// ── EVENT LISTENERS
──────────────────────document.getElementById("searchInput").addEventListener("input",
updateDisplay);document.getElementById("filterSelect").addEventListener("change", updateDisplay);// ── INITIALIZE
──────────────────────────updateDisplay();
```

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